Problem 1: Estonian Word Forms

Given below are some groups of words in Estonian[1] and their corresponding English translations. Each row contains words from the same group.

jooksmine	the action of running	jooksma	to run	jookslema	to run about
pikk	long	pikkus	length	pikkenema	to lengthen
lendus	flight	lendama	(1)	(A)	to flutter
(B)	to study	(C)	student	õpimine	(2)
suur	(3)	(D)	greatness	suurenema	to enlarge
voolav	fluid	(E)	fluidity	(F)	the action of flowing
välkatama	to flash once	välkama	(4)	välklema	to flicker
(G)	to know	(H)	science	teadlane	scientist
kangema	(5)	kangus	strength	(I)	hero
jumalasu	god/goddess	jumalasus	(6)	jumalasulane	priest
toukama	(7)	touklema	to jostle	toukamine	the action of pushing
selge	(8)	(J)	clarity	selgenema	(9)
(K)	abstinence	karsklane	abstainer	(L)	the action of abstaining
peen	(10)	(M)	finesse	peenenema	to refine
klõpsatama	to click once	klõpsma	to click	klõps	(11)
praksatama	to crack once	praksuma	(12)		
rändama	(13)	(N)	to wander		
(O)	honey	(P)	honeybee		
paks	thick	(Q)	to thicken		

Assignment 1: (12 marks)

Fill in the blank cells (A-Q) with Estonian words and the blank cells (1-13) with English words.

Assignment 2: (3 marks)

Given an English phrase, how would you form the corresponding Estonian word?

– Antara Raaghavi Bhattacharya and Aparna Gupte

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[1]The Estonian language is a Uralic language of the Finnic branch spoken in Estonia. It is the official language of Estonia, spoken natively by about 1.1 million people.

Problem 2: Stressed Manobo

Given below are some words in Sarangani Manobo[2]. Each word has the location of stress (i.e. emphasis) marked with ˈ at the beginning of the stressed syllable.

ˈbaso, deˈitek, meneˈnoo, meˈama, beˈgas, leˈkat, ˈotaw,
eteˈbay, bineleˈsan, ˈdeget, ˈbenget, miˈneles, ˈikan, ˈdoen

Assignment 1: (10 marks)

Mark the stress in the following Sarangani Manobo words with ˈ at the beginning of the stressed syllable:

(a) mengabat
(b) migbasa
(c) iselem
(d) benal
(e) tadon
(f) belegkong
(g) mola
(h) medaet

Assignment 2: (5 marks)

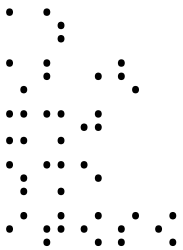
Given a new word in Sarangani Manobo, how would you identify the syllables in the word? Once you have identified the syllables, how would you determine which syllable is stressed?

– Saujas Vaduguru

Problem 3: Tenji Braille

Here are five Japanese words and their corresponding Tenji Braille[3] transliterations:

asa	morning
kaigi	meeting
fukuro	pocket
sakura	cherry blossom
otokonoko	boy



[2]Sarangani Manobo is one of the languages comprising the Manobo subgroup within the Philippine branch of theAustronesian language family. It is spoken by an estimated twenty thousand people living on the east coast of the Sarangani Peninsula and another ten thousand people living on the San Agustin Peninsula of southern Mindanao.

[3]Japanese Braille is the braille script of the Japanese language. It is based on the original braille script, though the connection is tenuous. In Japanese it is known as *tenji*, literally “dot characters”.

Here are some more Japanese words and their Tenji Braille transliterations, in no particular order:

1.	ikimono	living creatures	A.
2.	nezumi	mouse	B.
3.	sora	sky	C.
4.	shinzo	heart	D.
5.	ushiro	behind	E.
6.	hi	fire	F.
7.	kinoko	mushroom	G.
8.	mugi	wheat	H.
9.	bangohan	supper	I.
10.	mame	beans	J.

Assignment 1: (10 marks)

Match the Japanese words (1-10) to their Tenji Braille transliterations (A-J).

Assignment 2: (5 marks)

Transliterate the following words to Tenji Braille:

i.	geri	diarrhea
ii.	kirin	giraffe
iii.	saba	mackerel
iv.	hotaru	firefly
v.	atsumaru	to gather

– Apoorva Bhagwat and Sandipan Dandapat

Problem 5: Bahinemo

Here are the first ten multiples of a number less than 10 written in Bahinemo[4] in ascending order:

(a) həl baɸú həl ɸənbí husí

(b) həl baɸú həl baɸú wá ɸənbí gədà

(c) ímà tʰú hwəsə́ dəbátʰà

(d) ímà tʰú hwəsə́ yemú həl baɸú həl ɸənbí huməlí

(e) ímà tʰú hwəsə́ yemú həl baɸú həl baɸú wá baɸú

(f) ímà tʰú hwəsə́ husí husí

(g) ímà tʰú hwəsə́ husí həl baɸú həl ɸənbí gədà

(h) ímà tʰú hwəsə́ husí həl baɸú həl baɸú wá baɸú wá ɸənbí dəbátʰà

(i) ímà tʰú hwəsə́ huməlí huməlí

(j) ímà tʰú hwəsə́ huməlí həl baɸú həl baɸú

Bahinemo speakers often use a shortened form for some numbers. This, however, can lead to some ambiguity: a Bahinemo numeral in its shortened form can be interpreted as two different numbers.

Assignment 1: (4 marks)

Exactly one of the numbers given above is in its shortened form. All the other numbers are in the long form. In this case, the shortened numeral has only one interpretation, so there is no ambiguity. Which number is this, and what is its Hindu-Arabic equivalent? What is its Bahinemo long form? How is the shorthand form of a numeral derived from the long form?

Assignment 2: (8 marks)

Give all possible translations of the following Bahinemo numerals to Hindu-Arabic numbers. Some of these numerals may be in an ambiguous shortened form.

(a) ímà tʰú hwəsə́

(b) ímà tʰú hwəsə́ həl baɸú həl ɸənbí huməlí

(c) həl baɸú həl ɸənbí gədà

(d) ímà tʰú hwəsə́ həl baɸú həl baɸú həl baɸú həl baɸú wá ɸənbí husí

(e) ímà tʰú hwəsə́ həl baɸú

[4]Bahinemo is a Sepik language spoken in East Sepik Province, Papua-New Guinea. It is spoken in 4 villages, including in Gahom village of Tunap-Hunstein Rural LLG in East Sepik Province. Traditionally, Bahinemo speaking people rarely used large numbers in their daily life. Numbers above one hundred could in theory be correctly spoken, but the resulting phrase is so long and convoluted that hearers in the 1970s puzzled over them.

Assignment 3: (8 marks)

Translate the following Hindu-Arabic numerals to Bahinemo. When possible, give both the shorthand and the long form of the numbers in Bahinemo.

(a) 17

(b) 29

(c) 81

(d) 180

(e) 23

– Aparna Gupte

Problem 6: Altai-ed up

Here are some sentences in English and their corresponding Altaian[5] translations[6]:

мен кече ол атды сарайда тапганм.	I found that horse in the barn yesterday.
мен сенди айылда сакдым.	I waited for you at home.
бис ӧлӧнде ойнодыыс.	We played on the lawn.
ол кече бу самарады алган.	He/she received this letter yesterday.
олор кече кыр барган.	They went to a mountain yesterday.
бис кече ӱредичи школыпда сакганыс.	We waited for a teacher in school yesterday.
сен кече jӱсгенҥ.	You swam yesterday.
ӱредичи самара бичди.	The teacher wrote a letter.
зне кече бу бичикди кычырган.	The mother read this book yesterday.
уул кӧлди барды.	The boy went to the lake.
сен малды азырадын.	You fed the cattle.
бала самара турада тапды.	The child found a letter in the house.

[5]Altai is a Turkic language, spoken officially in the Altai Republic, Russia.

[6]а, о, у, ы are back/hard vowels and ӧ, ӱ, е, и are front/soft vowels. All other characters are consonants. Data has been simplified.

Assignment 1: (8 marks)

Translate the following English sentences to Altai:

(a) The mother found this book in the school.

(b) We played on the mountain yesterday.

(c) The boy fed a horse on the lawn.

(d) You wrote a letter yesterday.

Assignment 2: (8 marks)

Translate the following Altaian sentences to English:

(a) ол ол самарады кычырды.

(b) бала менди сакды.

(c) олор кече көлде jӱсген.

(d) мен кече школыпды барганм.

Assignment 3: (4 marks)

How are words or parts of words put together to form a valid Altaian sentence? What are their meanings and functions?

– Aparna Gupte

Problem 4: Kunimaipa

Sohopuho noho.	I went and then ate.
Sata noho.	I went and ate (same time).
Seŋipuho neŋi.	You went and then ate.
Sahapuho naha.	He went and then ate.
Hehopuho soho.	I stayed and then went.
Hehana soho.	He stayed and then I went.
Sahana noho.	He went and then I ate.
Sahana neŋi.	He went and then you ate.
Hemapuho soma.	I will stay and then go.
Hekepuho seke.	You will stay and then go.
Hepanepuho sapane.	He will stay and then go.
Heta naha.	He will stay and eat (same time).
Hemana seke.	I will stay and then you will go.
Hekena soma.	You will stay and then I will go.
Hepanena seke.	He will stay and then you will go.

Assignment 1: (5 marks)

Translate the following into English:

(a) Hehapuho saha

(b) Nata hema

(c) Heŋina saha

Assignment 2: (5 marks)

Translate the following into Kunimaipa:

(a) I will eat and then go.

(b) You will eat and go (at the same time).

(c) I will stay and then he will eat.

Assignment 3: (5 marks)

How are words or parts of words put together to form a valid Kunimaipa sentence? What are their meanings and functions?

– Zarana Sarda

Problem 6: Twisted Translations

Word embeddings are numeric representations of words as points in d-dimensional space. That is, a word can be represented using d numbers, where each number represents the coordinate of that number along a particular dimension.

For example, 2-dimensional embeddings are representations of words as points on a plane using a pair of numbers. The first number corresponds to the x-corrdinate of the point and the second number to the y-coordinate of the point.

These word embeddings are very useful representations of words for training machine learning models to perform natural langauge processing tasks. The embedding for a word is computed by looking at the word in a large corpus of text, and looking at what words occur in the context of that word.

Some machine translation models use embeddings of words in the source language to compute the embedding of the word in the target language, and then look up the word corresponding to that embedding. In this problem, we consider translating Hrangkhol to English[7].

[7]Hrangkhol or Hrangkhawl is a Kuki-Chin language spoken by the Hrangkhawl people mainly in Assam and Tripura states in India, with a minority living in Manipur and Mizoram.

Here are some English words, along with their embeddings in two dimensional space:[8]

(a) bull `[2, 2]`
(b) cow `[3, 3]`
(c) mare `[4, 5]`
(d) (female) duck `[3, -1]`
(e) son-in-law `[-5, 1]`
(f) daughter-in-law `[-4, 2]`
(g) grandson `[-6, 3]`

Here are some Hrangkhol words, along with their embeddings in two dimensional space:

(i) vôkchal `[-4, 6]`
(ii) vôkpui `[-6, 8]`
(iii) sakorchal `[-8, 6]`
(iv) arkhong `[4, 2]`
(v) arpui `[2, 4]`
(vi) vatôkhong `[4, 4]`
(vii) vaikikhong `[8, 4]`
(viii) naipa `[-4, -12]`
(ix) nainu `[-6, -10]`
(x) tunu `[-8, -10]`

The English words corresponding to (i-x) in **random** order are:[9]

(A) male parrot
(B) hen
(C) sow
(D) daughter
(E) boar
(F) granddaughter
(G) drake
(H) stallion
(I) son
(J) rooster

Assignment 1: (10 marks)

Match the Hrangkhol words (i-x) to their English translations (A-J).

Assignment 2: (3 marks)

Write down the word embeddings for the Hrangkhol translations of the words (a-g).

Assignment 3: (4 marks)

Write down the word embeddings for the English translations of the words (i-x).

[8]A bull is an adult male cow, and a mare is an adult female horse.
[9]A sow is an adult female pig and a boar is an adult male pig. A stallion is an adult male horse. A drake is an adult male duck. A rooster is an adult male chicken.

Assignment 4: (3 marks)

Translate the following words to Hrangkhol:

(I) female parrot

(II) he-goat (root of goat is kêl)

(III) old woman (root of old person is tar)

– Saujas Vaduguru